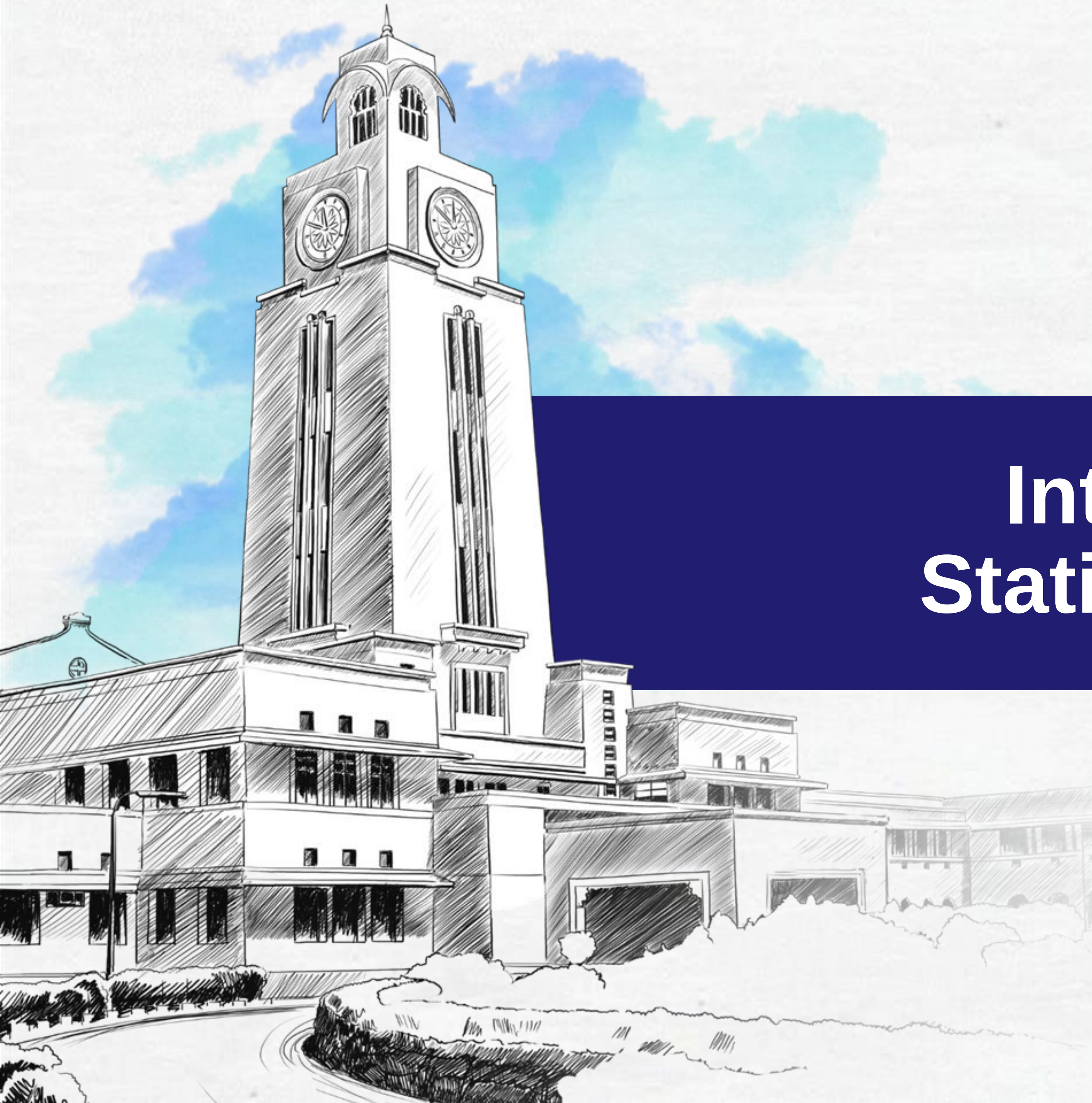

BITS Pilani
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Introduction to Statistical Methods

ISM Team

Session 11
(17th January 2026)

Correlation and Regression

Example 1

Suppose you define the time to get ready as the time (rounded to the nearest minute) from when you get out of bed to when you leave your home. You collect the times shown below for 10 consecutive work days:

Day	1	2	3	4	5	6	7	8	9	10
Time (min)	39	29	43	52	39	44	40	31	44	35

How do you summarize the above data?

Example 2

The owner of a showtime movie theatre wishes to predict weekly gross revenue based on advertising expenditures. The sample data are as follows:

Sl. No	1	2	3	4	5	6	7	8
Weekly gross revenue (‘1000s)	96	90	95	92	95	94	94	94
Television advertising (‘1000s)	5.0	2.0	4.0	2.5	3.0	3.5	2.5	3.0
Newspaper advertising (‘1000s)	1.5	2.0	1.5	2.5	3.3	2.3	4.2	2.5

Is summarization of the above data enough? Or do we need to look beyond just summerization?

In example 1, as the data relates to a single variable, only summarization with descriptive statistics is enough. But in example 2, as there are three variables, besides summarization by descriptive statistics, it is also required to find out whether, there is relation between the variables.

In summary statistics, one important measure besides mean is variance (or standard deviation)

Covariance

- Likewise, in the pursuit of finding the relation between two or more variables, the statistical technique used for studying the existence and extent of relationship between the variables is through Covariance

$$Cov\,(X,Y) = E(X - \bar{X})(Y - \bar{Y})$$

- Covariance signifies the direction of the linear relationship between the variables. By direction we mean if the *variables* are directly proportional or inversely proportional to each other. (Increasing the value of one variable might have a positive or a negative effect on the value of the other variable).

Covariance

- The values of covariance can be any number between the two opposite infinities. Also, it’s important to mention that covariance only measures how two variables change together, not the dependency of one variable on another one.
- The upper and lower limits for the covariance depend on the variances of the variables involved. These variances, in turn, can vary with the scaling of the variables. Even a change in the units of measurement can change the covariance. Thus, covariance is only useful to find the direction of the relationship between two variables and not the magnitude.

Covariance

- If X_1 and X_2 are two variables, the covariance between X_1 and X_2 is defined as

$$\mathrm{Cov}(X_1, X_2) = \frac{\sum_{i=1}^{n}(X_1 - \overline{X}_1)(X_2 - \overline{X}_2)}{n-1}$$

- Even though the covariance explains the relationship between two or more variables, the units the measurement of variable is attached with the value of the covariance. If they have different unit of measurements, based on the sign it is possible to say whether they are related or not, but the degree (strength) of relationship is not possible to decide based on the value of covariance due to different units of measurement.

Covariance

- If $X_1, X_2, \ldots, X_n$ are n variables, then the Covariance between these n variables is the Variance – Covariance matrix given by

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & . & . & . & \sigma_{1n} \\ \sigma_{21} & \sigma_{22} & . & . & . & \sigma_{2n} \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ \sigma_{n1} & \sigma_{n2} & . & . & . & \sigma_{nn} \end{bmatrix}$$

where

$$\sigma_{ij} = \begin{cases} \text{Variance, for } i = j \\ \text{Covariance, for } i \neq j \end{cases}$$

All diagonal elements in the variance-covariance matrix are variances and off diagonal elements are covariances.

Σ is a symmetric matrix.

Covariance

Example: Find the covariance between age (months) and weight (kgs) for the following data

Age (years)	7	6	8	5	6	9
Weight (kgs)	12	8	12	10	11	13

Covariance

Age (yrs) (X)	Weight (kgs) (Y)	$(X-\overline{X})$	$(Y-\overline{Y})$	$(X-\overline{X})(Y-\overline{Y})$
5	20	-2	-3	6
4	18	-3	-5	15
7	23	0	0	0
10	32	3	9	27
8	26	1	3	3
8	25	1	2	2
$\sum X = 42$	$\sum Y = 144$			$\sum (X-\overline{X})(Y-\overline{Y}) = 53$

$$\mathrm{Cov}(X, Y) = \frac{53}{5} = 10.6 \text{ years.kgs} \rightarrow \text{what is the interpretation of this value?}$$

Covariance

Covariance based on Probability distribution.

If X and Y are two random variables with probability mass function p(x)/ probability density function f(x), then the covariance between X and Y

$$\mathbf{Cov\ (X, Y) = E[(X-\mu_X)(Y-\mu_Y)] = E(XY) - E(X)E(Y)}$$

where $E(X) = \mu_X$ and $E(Y) = \mu_Y$ are the means of X and Y.

Covariance

- Application of Covariance.

- 1. Multivariate linear regression:

- If there a dependent variable and n independent variables, to find out the relationship between these (n+1) variables, a variance – covariance matrix will be helpful.

- 2. Time series:

- In the given series of data over the time, it helps in finding the autocorrelation at different time lags.

- For example, if $X_{t_1}, X_{t_2}, \ldots, X_{t_n}$ denote the time series data over a given period, then the autocovariance at lag h is given by $\gamma_X(t+h, t) = Cov(X_{t+h}, X_t)$

Covariance

- For example, if $X_{t_1}, X_{t_2}, \ldots, X_{t_n}$ denote the time series data over a given period, then the auto-covariance.
- The auto-covariance at lag h is then given by

$$\hat{\gamma}(h) = \frac{1}{n}\sum_{i=1}^{n-|h|}(X_{t+|h|} - \overline{X})(X_t - \overline{X}), -n < h < n.$$

- With the help of auto-covariance, the autocorrelation is given by

$$r_k = \frac{\sum_{t=k+1}^{T}(y_t - \overline{y})(y_{t-k} - \overline{y})}{\sum_{t=1}^{T}(y_t - \overline{y})^2}$$

Application of Covariance

2. Principal Component Analysis (PCA):

- PCA is simply described as "diagonalizing the covariance matrix". What does diagonalizing a matrix mean in this context? It simply means that we need to find a non-trivial linear combination of our original variables such that the covariance matrix is diagonal.
- When this is done, the resulting variables are uncorrelated, i.e. independent. The computation of the linear combinations of the original variables, called the 'principal components'.
- Let A be a variance – covariance matrix and X be a vector of variables the $Ax = \lambda x$, where λ is a scalar.

Application of Covariance

- Calculate the eigenvalues and the corresponding eigenvectors of the covariance of the matrix A. If an eigenvalue computed in PCA is large, then the corresponding eigenvector (principal component) is important in describing the underlying data dependencies.
- If the eigenvalue is near zero, that principal component is relatively unimportant and the data depend primarily on fewer components than there are measurement types. We merely need the machinery of eigenvectors and covariance matrices in order to find linear combinations of our original variables that are uncorrelated.
-

Correlation coefficient

- To overcome the issue related to Covariance, Karl Pearson's suggestion was to divide the covariance of the variables by their respective standard deviations. The ratio of covariance to the product of standard deviations is called Karl Pearson's product moment correlation, which is free of unit of measurement.
- Karl Pearson's product moment correlation returns a numerical value, popularly known as **Coefficient of Correlation**, denoted by **'r'**
- Correlation coefficient is a statistic that assesses the **strength** and **direction of relationship** of two continuous variables

Correlation coefficient

- Mathematically the following measures are used to find out the degree of relationship between the variables
  - Karl Pearson’s product moment correlation
  - Spearman’s rank correlation
- The numerical value obtained by the above methods is called correlation coefficient
- Karl Pearson’s product moment correlation or product moment correlation is also called simple correlation coefficient, denoted by ‘r’
- It measures the nature and strength of relationship between two variables of the quantitative type

Karl Pearson's Correlation coefficient

- Formula for computing Karl Pearson's correlation coefficient (r) is standardised covariance (unit less)

$$r = \frac{\text{Covariance}(x, y)}{\sqrt{\text{Var}(x)}\sqrt{\text{Var}(Y)}}$$

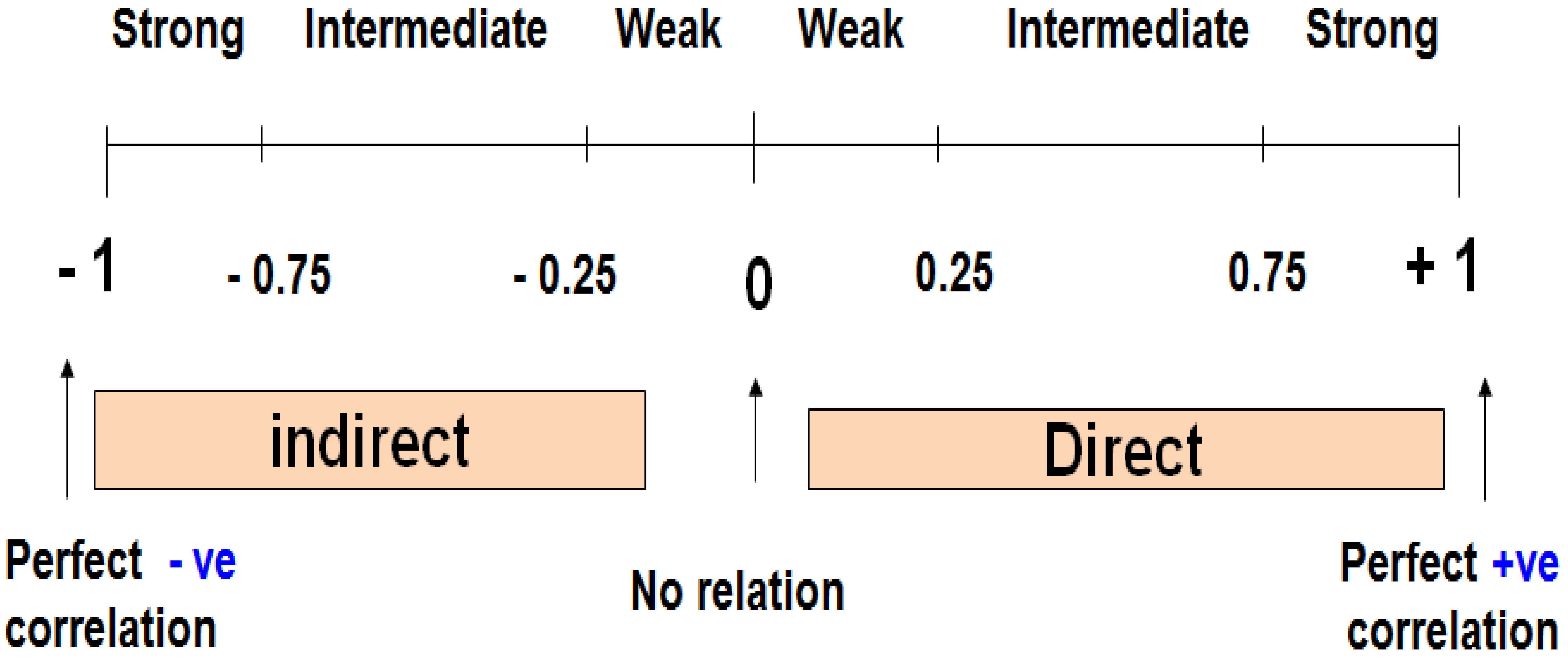
$$r = \frac{\dfrac{\sum_{i=1}^{n}(X - \overline{X})(Y - \overline{Y})}{n - 1}}{\sqrt{\dfrac{\sum_{i=1}^{n}(X - \overline{X})^2}{n-1}}\sqrt{\dfrac{\sum_{i=1}^{n}(Y - \overline{Y})^2}{n-1}}} = \frac{\sum_{i=1}^{n}(X - \overline{X})(Y - \overline{Y})}{\sqrt{\sum_{i=1}^{n}(X - \overline{X})^2}\sqrt{\sum_{i=1}^{n}(Y - \overline{Y})^2}}$$

$$r = \frac{n\sum XY - \left(\sum X\right)\left(\sum Y\right)}{\sqrt{n\sum X^2 - \left(\sum X\right)^2}\sqrt{n\sum Y^2 - \left(\sum Y\right)^2}}$$

Karl Pearson's Correlation coefficient

- The <u>sign</u> of 'r' denotes the nature of relationship
- The <u>value</u> of 'r' denotes the strength of relationship.
- If the sign is +ve, this means the relation is direct (an increase in one variable is associated with an increase in the other variable and a decrease in one variable is related with a decrease in the other variable).
- If the sign is -ve this means an inverse or indirect relationship (an increase in one variable is related with a decrease in the other).
- The value of r ranges between (-1) and (+1), i.e., $-1 \leq r \leq +1$
- The value of r denotes the strength of the relationship as illustrated by the following diagram.

Karl Pearson’s Correlation coefficient



Karl Pearson’s Correlation coefficient

- $r = -1 \Rightarrow$ Perfect negative correlation
- -0.99 to $-0.76 \Rightarrow$ Strong negative correlation
- -0.75 to $-0.26 \Rightarrow$ Intermediate (Moderate) negative correlation
- -0.25 to $0 \Rightarrow$ Weak negative correlation
- $r = 0 \Rightarrow$ Zero correlation
- 0 to $0.25 \Rightarrow$ Weak positive correlation
- 0.26 to $0.75 \Rightarrow$ Intermediate (Moderate) positive correlation
- 0.76 to $0.99 \Rightarrow$ Strong positive correlation
- $r = +1 \Rightarrow$ Perfect positive correlation

Karl Pearson’s Correlation coefficient

Serial No	Age (yrs)	Weight (Kg)
1	7	12
2	6	8
3	8	12
4	5	10
5	6	11
6	9	13

Karl Pearson's Correlation coefficient

Sl no.	Age (yrs) (X)	Wt (Kg) (Y)	XY	X^2	Y^2
1	7	12	84	49	144
2	6	8	48	36	64
3	8	12	96	64	144
4	5	10	50	25	100
5	6	11	66	36	121
6	9	13	117	81	169
Total	$\sum X=41$	$\sum Y=66$	$\sum XY= 461$	$\sum X^2= 291$	$\sum Y^2= 742$

Karl Pearson's Correlation coefficient

$$r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{n\sum X^2 - (\sum X)^2}\sqrt{n\sum Y^2 - (\sum Y)^2}}$$

$$r = \frac{6 * 461 - 41 * 66}{\sqrt{6 * 291 - (41)^2}\sqrt{6 * 742 - (66)^2}}$$

r = 0.759 (Positive (Direct) strong correlation)

Karl Pearson's Correlation coefficient

Example: Relationship between anxiety and test Scores

Anxiety (X)	Test score (Y)	X^2	Y^2	XY
10	2	100	4	20
8	3	64	9	24
2	9	4	81	18
1	7	1	49	7
5	6	25	36	30
6	5	36	25	30
$\sum X = 32$	$\sum Y = 32$	$\sum X^2 = 230$	$\sum Y^2 = 204$	$\sum XY = 129$

Karl Pearson's Correlation coefficient

$$r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{n\sum X^2 - (\sum X)^2}\sqrt{n\sum Y^2 - (\sum Y)^2}}$$

$$r = \frac{6*129 - 32*32}{\sqrt{\left(6*230 - (32)^2\right)\left(6*204 - (32)^2\right)}} = \frac{774 - 1024}{\sqrt{(356)(200)}} = -0.94$$

r = - 0.94 (Negative (Indirect) strong correlation)

Fig 1: Perfect negative correlation: r = – 1

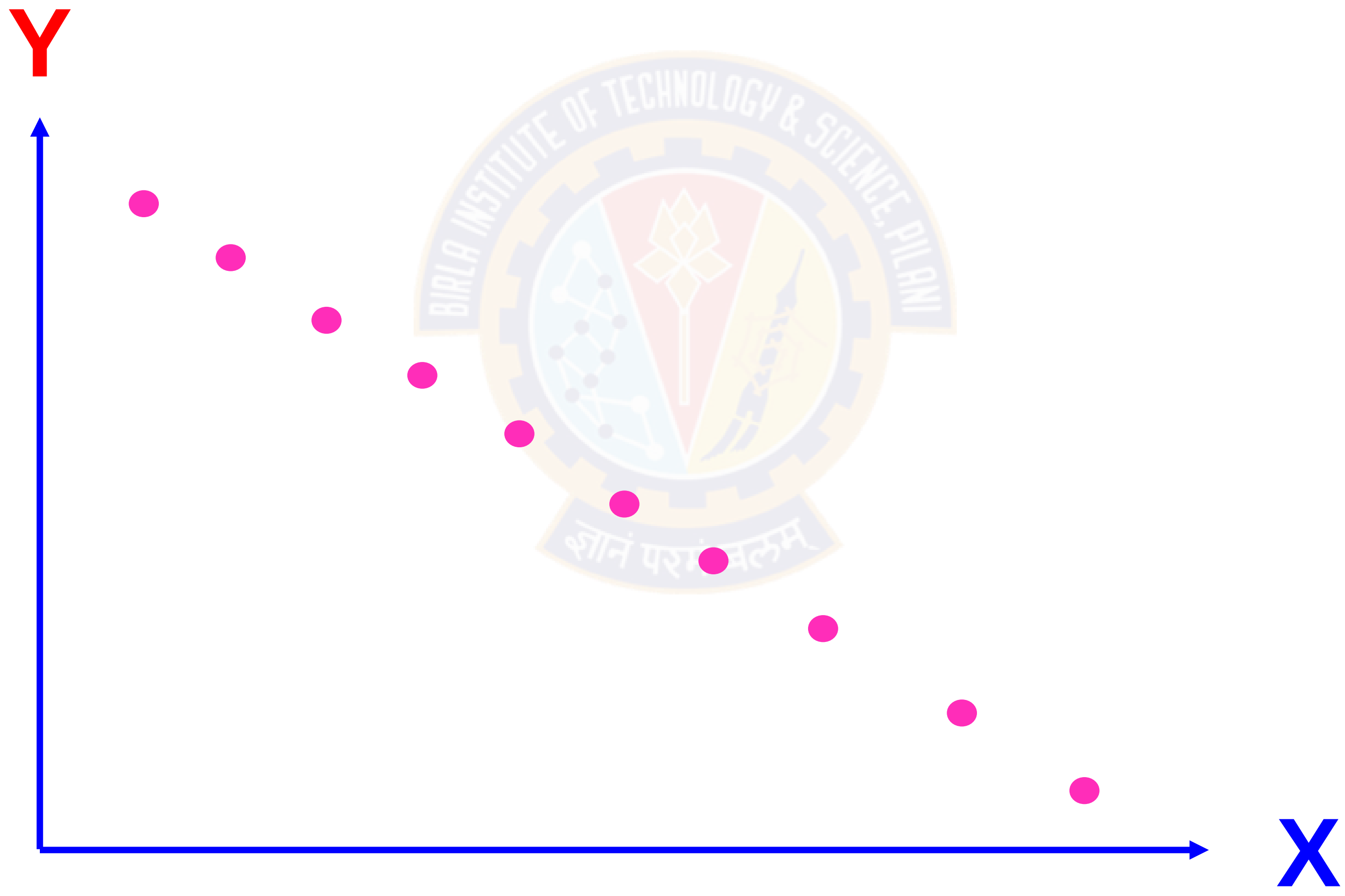


Fig 2: Strong negative correlation: $-0.99 \leq r < -0.75$

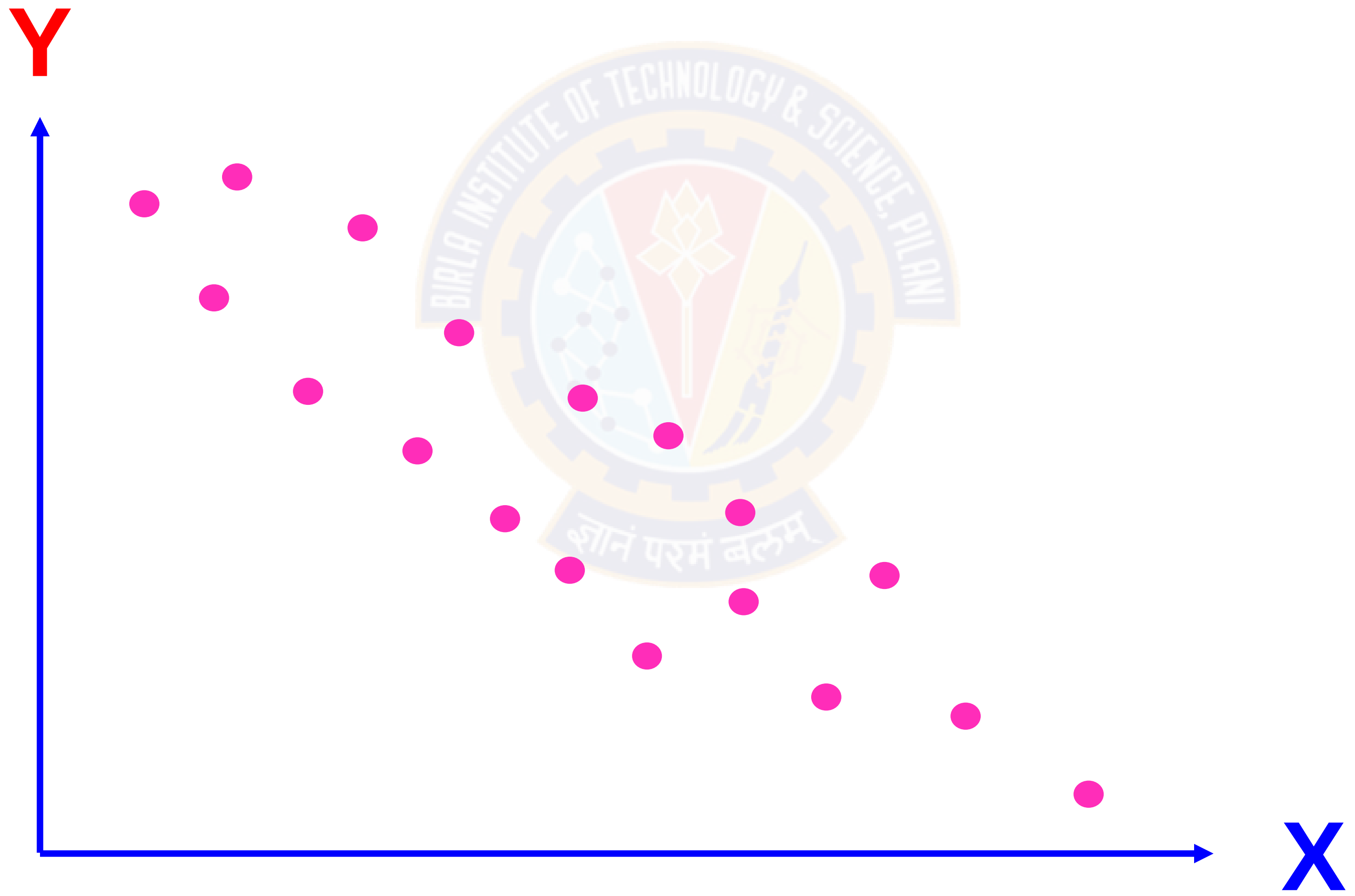


Fig 3: Moderate negative correlation: $-0.75 \le r < -0.25$

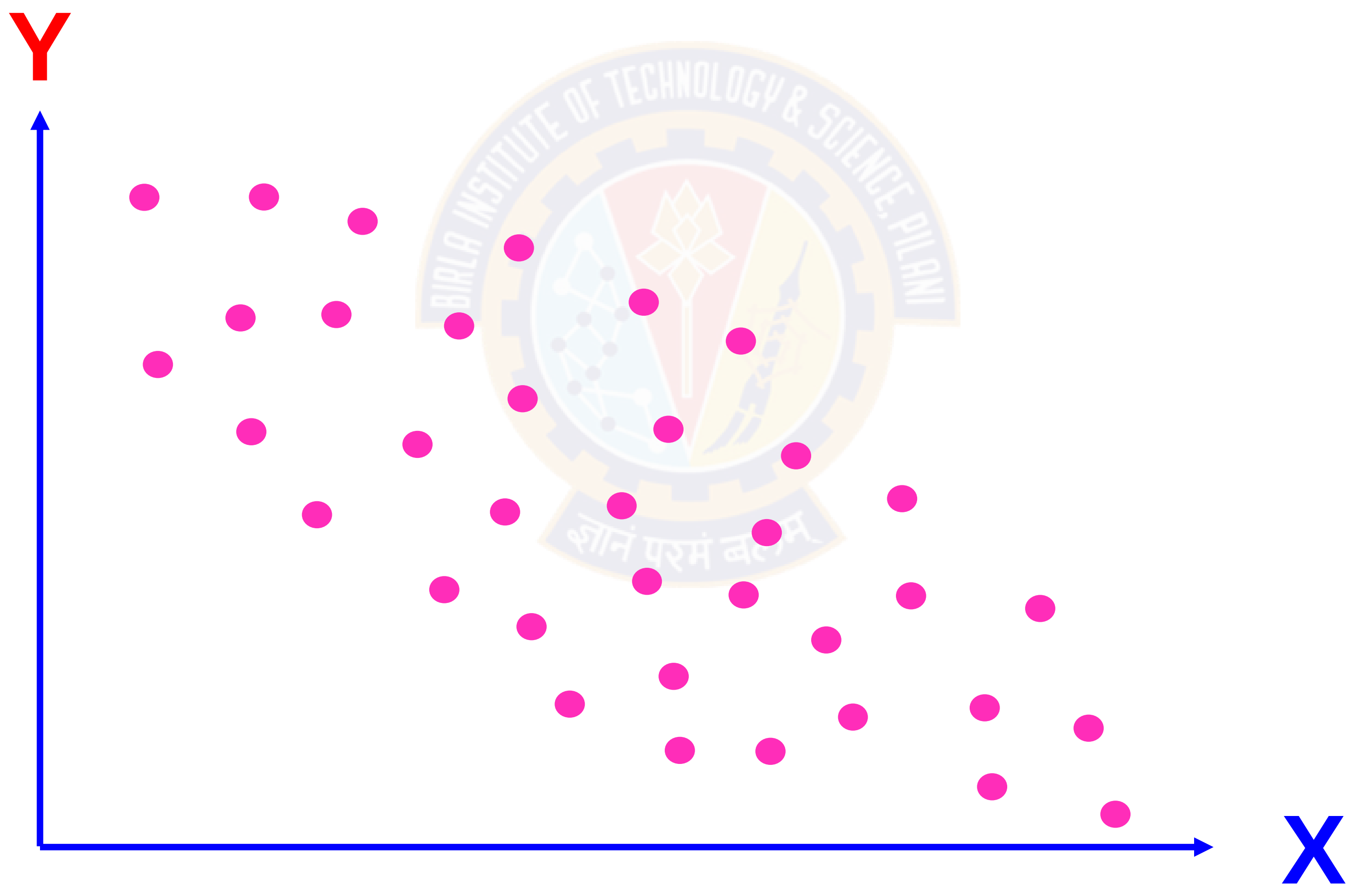


Fig 4: Weak negative correlation: $-0.25 \leq r < 0$

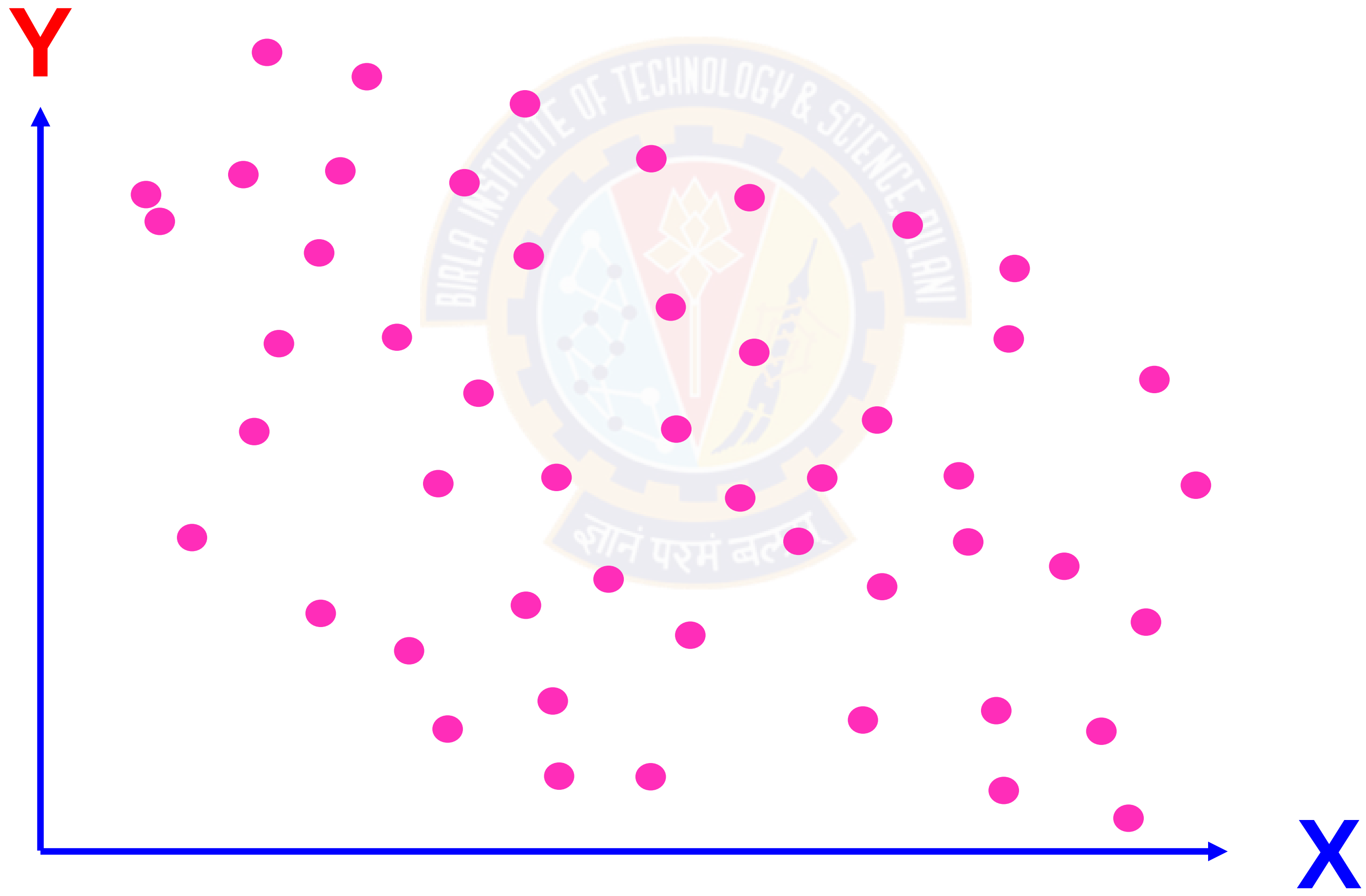


Fig 5 (a): Zero correlation: r = 0

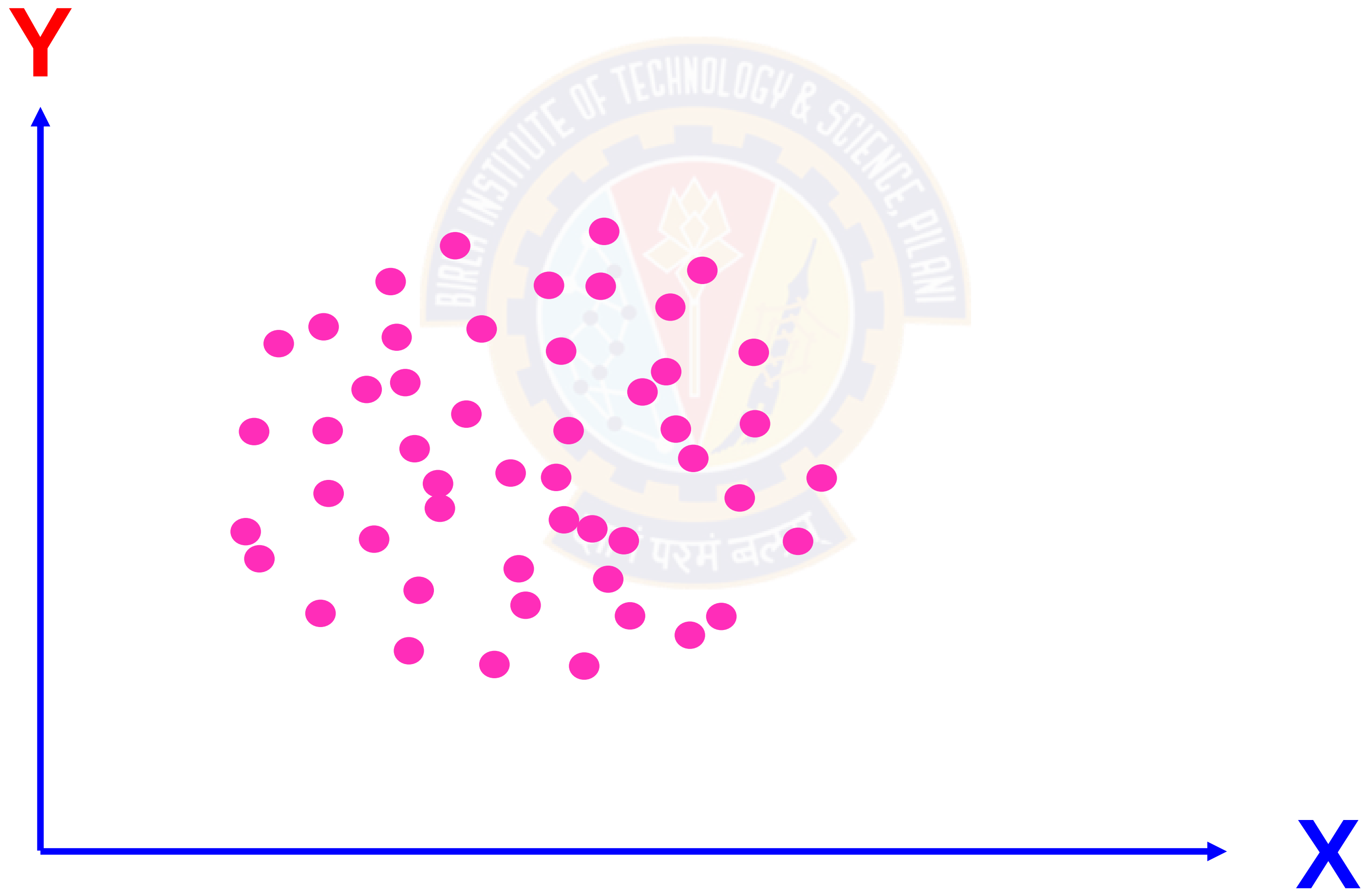


Fig 5 (b): Zero correlation: r = 0

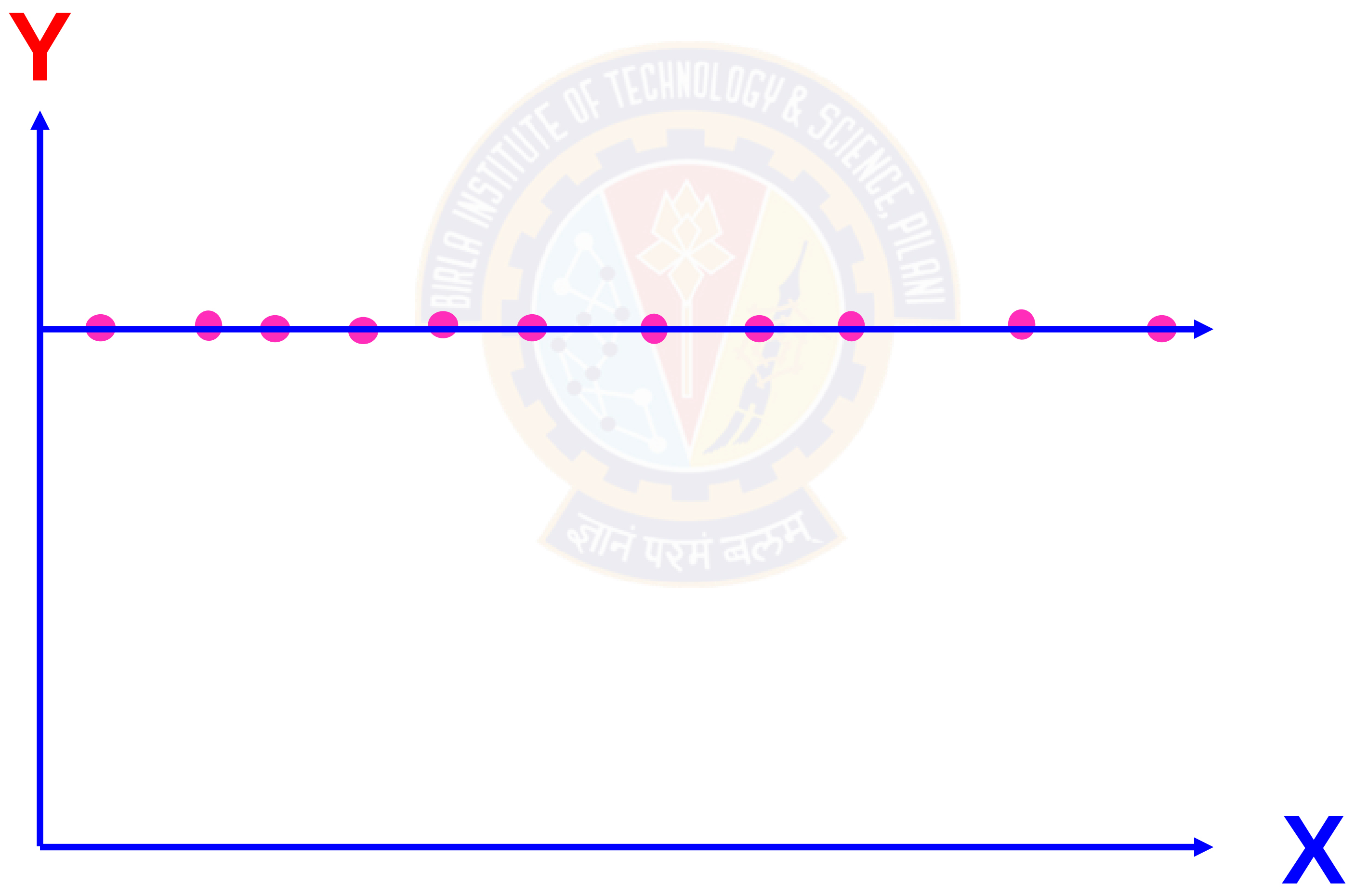


Fig 5 (c): Zero correlation: r = 0

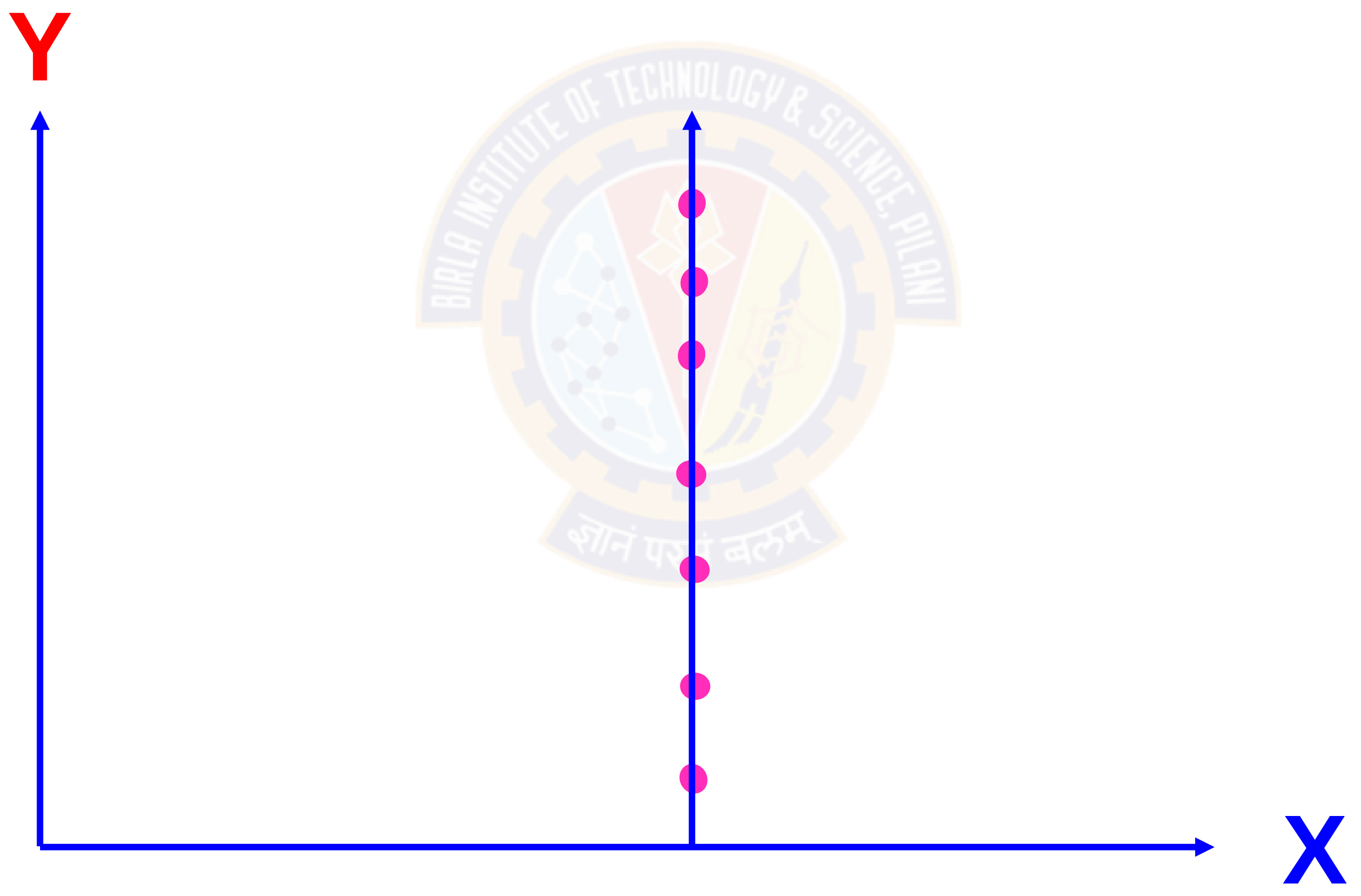


Fig 6: Weak positive correlation: $0 < r \le 0.25$

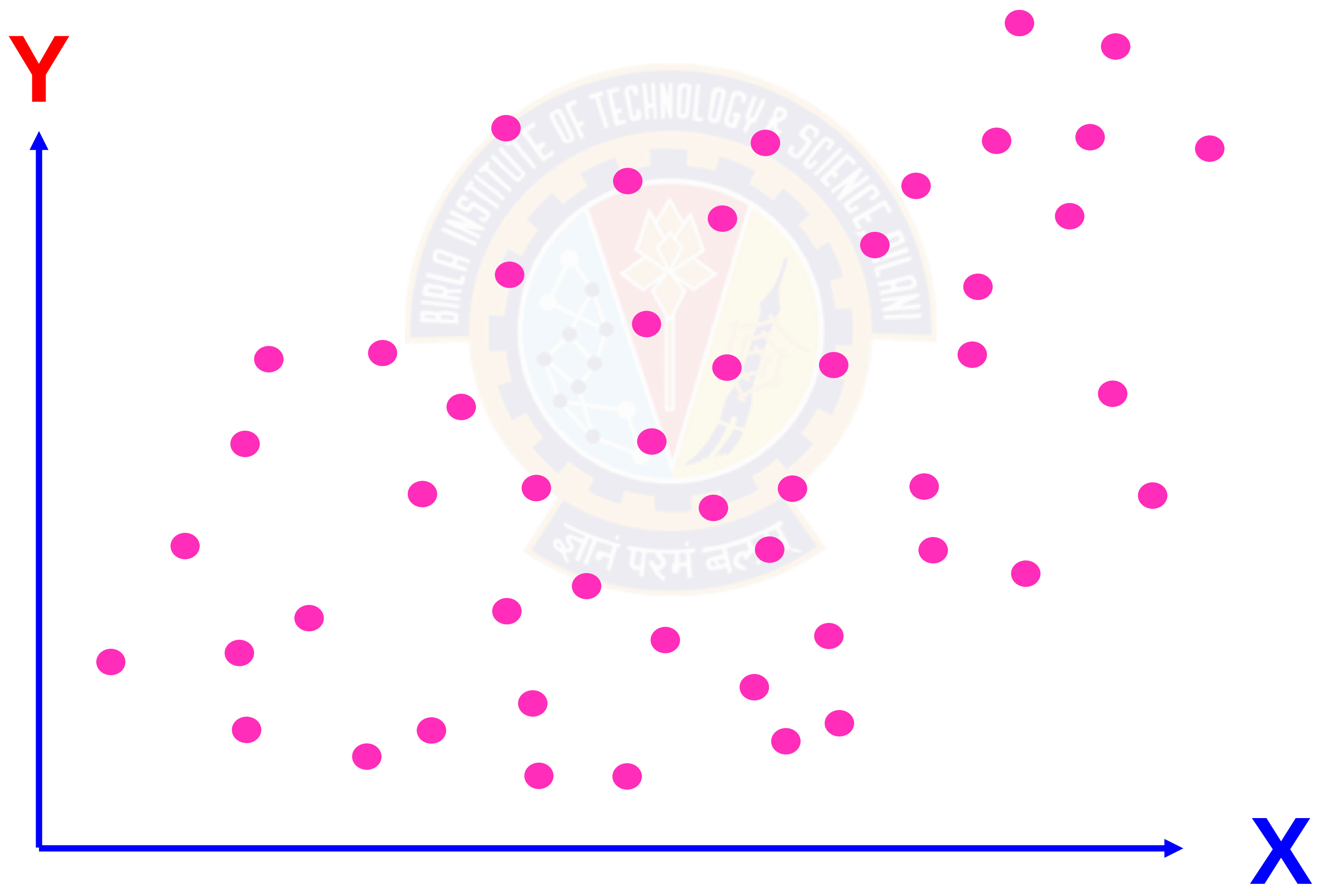


Fig 7: Moderate positive correlation: $0.25 < r \leq 0.75$

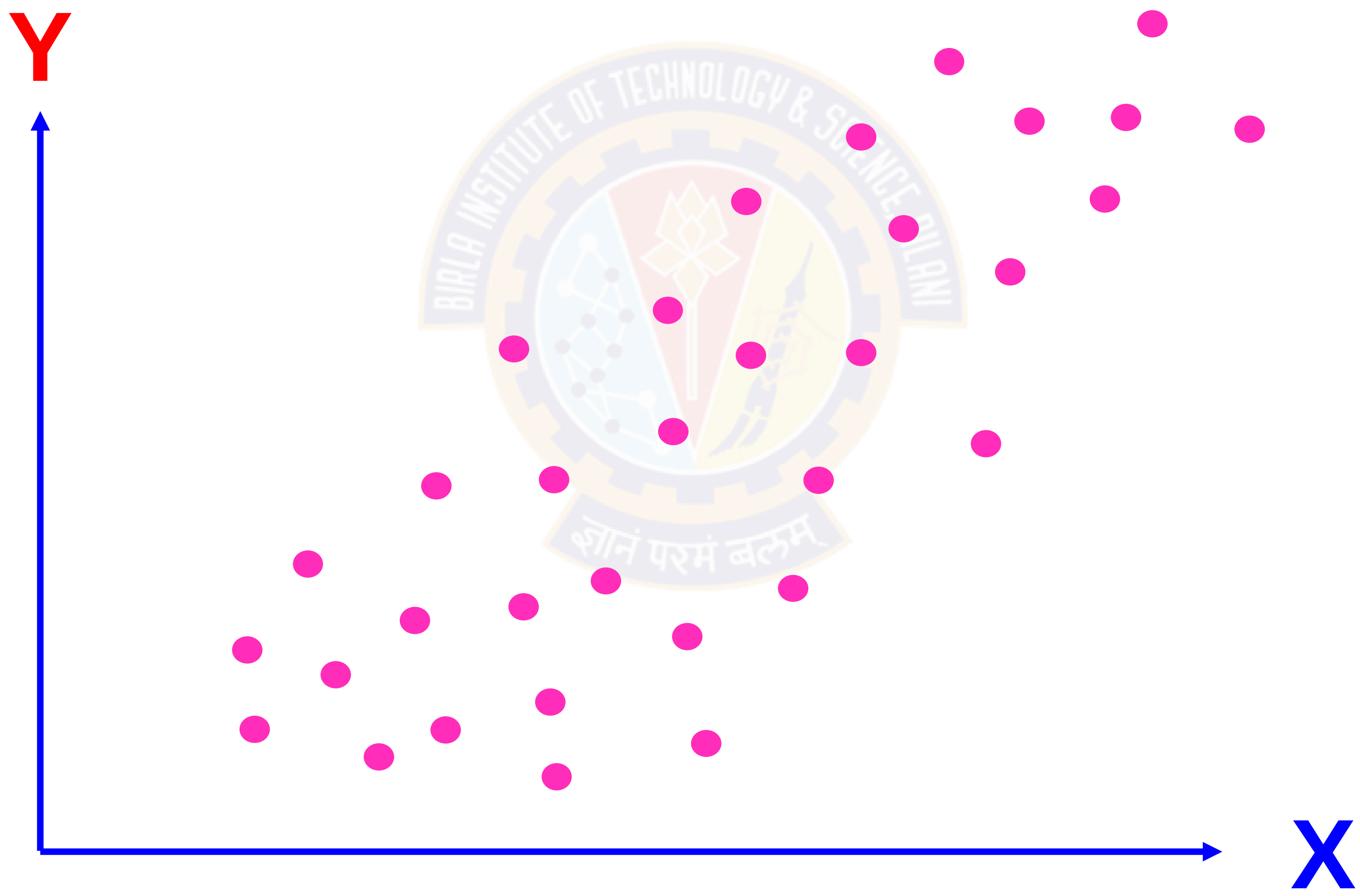


Fig 8: Strong positive correlation: $0.75 < r \leq 0.99$

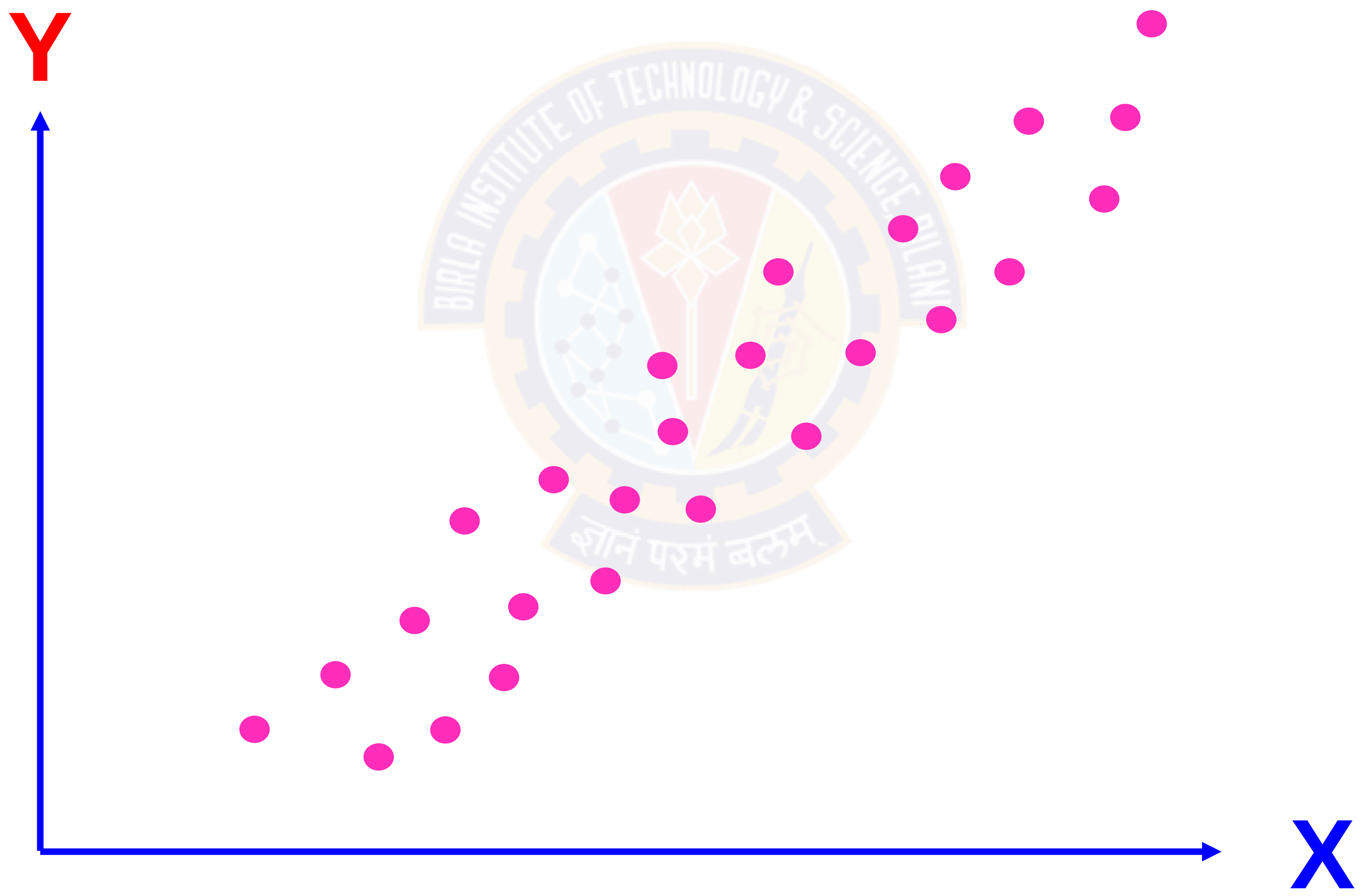
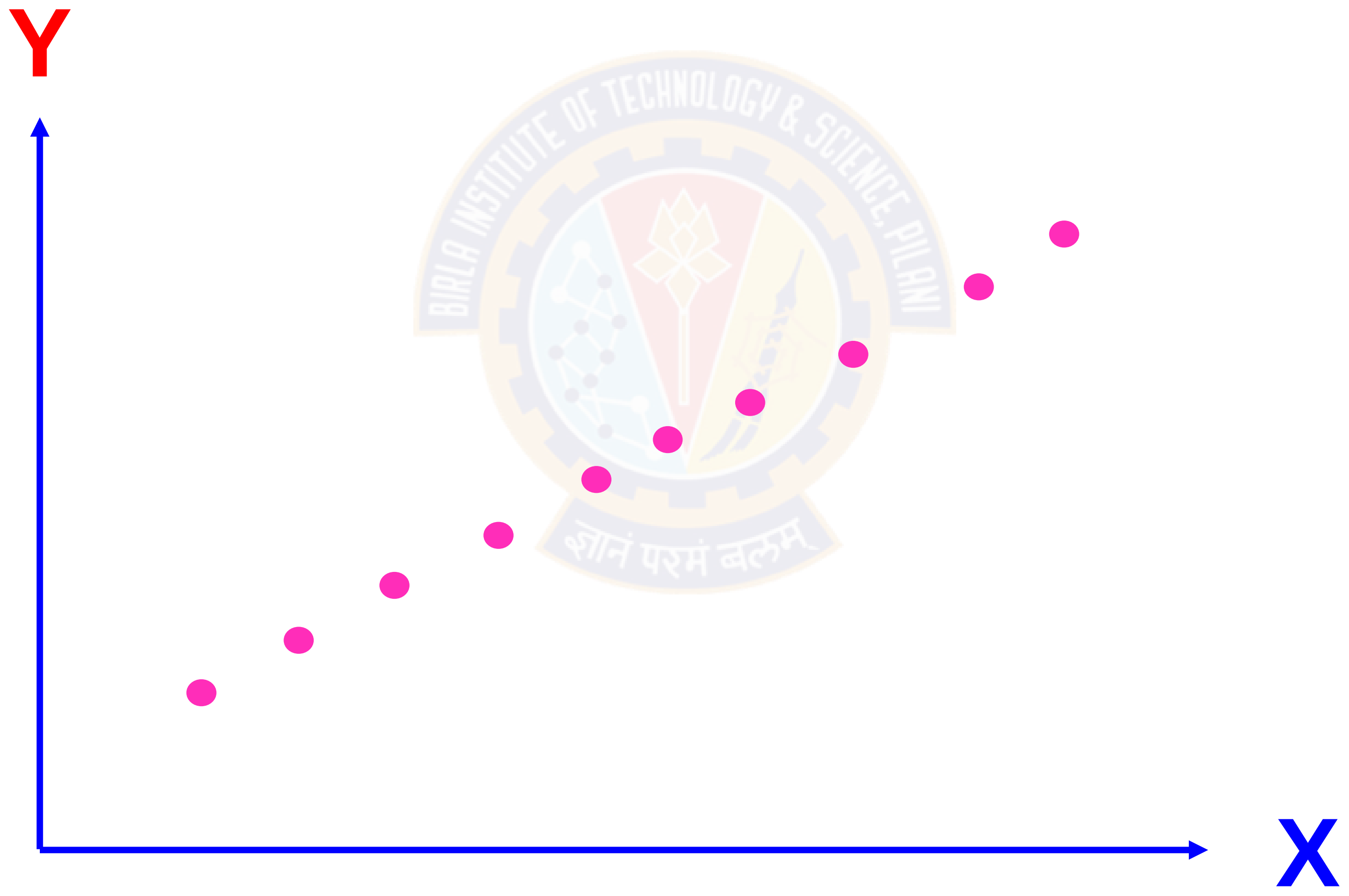


Fig 9: Perfect negative correlation: r = + 1



Regression Analysis

- Weekly gross revenue (Y) = Television advertising (X)

  That is **$Y = X$**
- But TV advertising will not be constant all the week. By adding a weighting factor to X to account for variability, the model becomes

  $Y = bX$
- Weekly gross revenue may not depend only on TV ad, but other factors may also act in increasing the revenue. Now, by adding 'a' as an anchor point, the modified model will be

Regression Analysis

$$Y = a + bX$$

While making measurements on Y and X, the possibility of error due to randomness has to be considered, thereby the full the model becomes

$$Y = a + bX + e$$

This model is called as simple linear regression model which can be used for predicting the weekly gross revenue based on Television advertisement.

Simple Linear Regression

Relationships:

- **Function**: a mathematical relationship enabling us to predict what values of one variable (Y) correspond to given values of another variable (X).
- Y: is referred to as the *dependent variable*, the *response variable* or the *predicted variable*.
- X: is referred to as the *independent variable*, the *explanatory variable* or the *predictor variable*.

Model Assumptions about the error term Ɛ

1. The error Ɛ is a random variable with mean of zero.
2. The variance of Ɛ, denoted by σ^2, is the same for all values of the independent variable.
3. The values of Ɛ are independent.
4. The error Ɛ is a normally distributed random variable.

Simple Linear Regression

Least Squares (L-S): Minimize squared error.

$$\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2 = \sum_{i=1}^{n} \hat{\varepsilon}_i^2$$

$$\sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} (Y_i - a - bX_i)^2$$

$$\frac{\partial}{\partial a}\sum_{i=1}^{n} \varepsilon_i^2 = \frac{\partial}{\partial a}\sum_{i=1}^{n} (Y_i - a - bX_i)^2 = 0$$

$$\sum_{i=1}^{n} Y_i = na - b\sum_{i-1}^{n} X_i \ldots\ldots\ldots\ldots(1)$$

Simple Linear Regression

Least Squares (L-S): Minimize squared error.

$$\sum_{i=1}^{n}\left(Y_i-\hat{Y}_i\right)^2=\sum_{i=1}^{n}\hat{\varepsilon}_i^2$$

$$\sum_{i=1}^{n}\varepsilon_i^2=\sum_{i=1}^{n}(Y_i-a-bX_i)^2$$

$$\frac{\partial}{\partial b}=\sum_{i=1}^{n}\varepsilon_i^{\,2}=\frac{\partial}{\partial b}\sum_{i=1}^{n}(Y_i-a-bX_i)^2=0$$

$$\sum_{i=1}^{n}\mathrm{Y_i}=\left(\mathrm{a}\sum_{i=1}^{n}\mathrm{X_i}-\mathrm{b}\sum_{i=1}^{n}\mathrm{X_i}\right)(-2X_i)=0$$

Simple Linear Regression

$$\frac{\partial}{\partial b}\sum_{i=1}^{n}\varepsilon_i^2 = \frac{\partial}{\partial b}\sum_{i=1}^{n}(Y_i - a - bX_i)^2 = 0$$

$$\sum_{i=1}^{n} Y_i = \left(a\sum_{i=1}^{n} X_i - b\sum_{i=1}^{n} X_i \right)(-2X_i) = 0$$

$$\sum_{i=1}^{n} X_i Y_i = a\sum_{i=1}^{n} X_i + b\sum_{i=1}^{n} X_i^{\,2} \ldots\ldots\ldots\ldots\ldots(2)$$

Equations (1) and (2) are called normal equations. Solving these as simultaneous equations, we get

Simple Linear Regression

$$\sum_{i=1}^{n} Y_i = na + b\sum_{i=1}^{n} X_i \ldots\ldots\ldots\ldots\ldots\ldots(1)$$

$$\sum_{i=1}^{n} X_i Y_i = a\sum_{i=1}^{n} X_i + b\sum_{i=1}^{n} X_i^{2} \ldots\ldots\ldots\ \ldots\ldots(2)$$

$$\hat{b} = \frac{n\sum_{i=1}^{n} X_i Y_i - \left(\sum_{i=1}^{n} X_i\right)\left(\sum_{i=1}^{n} Y_i\right)}{n\sum_{i=1}^{n} X_i^{2} - \left(\sum_{i=1}^{n} Xi\right)^2} = \boxed{\frac{\sum_{i=1}^{n} (X_i - \overline{X})(Y_i - \overline{Y})}{\sum_{i=1}^{n} (Xi - X)^2}}$$

$$\hat{a} = \overline{Y} - \hat{b}\overline{X}$$

The fitted regression model is

$$\hat{Y} = \hat{a} + \hat{b}X$$

The owner of a showtime movie theatre wishes to predict weekly gross revenue based on advertising expenditures. The sample data are as follows:

Sl. No	1	2	3	4	5	6	7	8
Weekly gross revenue ('1000s)	96	90	95	92	95	94	94	94
Television advertising ('1000s)	5.0	2.0	4.0	2.5	3.0	3.5	2.5	3.0

	Weekly gross revenue ('1000s) (Y)	Television advertising ('1000s) (X)	$x = (X - \bar{X})$	$y = (Y - \bar{Y})$	xy	x^2
	96	5.0	2.25	1.81	4.08	5.06
	90	2.0	-3.75	-1.19	4.45	14.06
	95	4.0	1.25	0.81	1.02	1.56
	92	2.5	-1.75	-0.69	1.20	3.06
	95	3.0	1.25	-0.19	-0.23	1.56
	94	3.5	0.25	0.31	0.08	0.06
	94	2.5	0.25	-0.69	-0.17	0.06
	94	3.0	0.25	-0.19	-0.05	0.06
Sum	750	25.5	0	0	**10.375**	**25.5**
Mean	**93.75**	**3.19**				

Simple Linear Regression

$$\hat{b} = \frac{\sum xy}{\sum x^2} = \frac{10.375}{25.5} = 0.407$$

$$\hat{a} = \bar{Y} - \hat{b} * \bar{X} = 93.75 - 0.407 * 3.15 = 92.45$$

$$\hat{Y} = \hat{a} + \hat{b} * X = 92.45 + 0.407X$$

When X = 4.5, $\hat{Y} = 92.45 + 0.407 * 4.5 = 94.28$

Simple Linear Regression

- Coefficient of Determination
- Relationship Among SST, SSR, SSE

$$SST = SSR + SSE$$

$$\sum(Y_i - \overline{Y})^2 = \sum(\hat{Y}_i - \overline{Y})^2 + \sum(Y_i - \hat{Y}_i)^2$$

where:

SST = Sum of Squares due to Total

SSR = Sum of Squares due to Regression

SSE = Sum of Squares due to Error

Simple Linear Regression

- Coefficient of Determination

$$r^2 = \frac{SSR}{SST} = \frac{\sum(\hat{Y} - \bar{Y})^2}{\sum(Y - \bar{Y})^2}$$

where:

SST = Sum of Squares due to Total

SSR = Sum of Squares due to Regression

Simple Linear Regression

Example

No. of TV advs (X)	No. of cars sold (Y)
1	14
3	24
2	18
1	17
3	27
2	20

Fit a regression line of Y on X and determine the coefficient of determination

Simple Linear Regression

	X	Y	x = (X-2)	y = (Y-20)	xy	x^2	$(Y-20)^2$	Y^ = 10 + 5X	$(Y\text{^} - 20)^2$
	1	14	-1	-6	6	1	36	15	25
	3	24	1	4	4	1	16	25	25
	2	18	0	-2	0	0	4	20	0
	1	17	-1	-3	3	1	9	15	25
	3	27	1	7	7	1	49	25	25
Total	10	100	0	0	20	4	114		100

Regression y on X

Coefficient of Determination

$$\hat{Y} = 10 + 5X$$

$$r^2 = \frac{SSR}{SST} = \frac{\sum(\hat{Y} - \bar{Y})^2}{\sum(Y - \bar{Y})^2} = \frac{100}{114} 0.8772$$

Simple Linear Regression

SL. No	X	Y	x = (X-2)	y = (Y-20)	xy	x^2	$(Y-20)^2$	Y^ = 10 + 5X	$(Y\text{\^{}} - 20)^2$
1	1	14	-1	-6	6	1	36	15	25
2	3	24	1	4	4	1	16	25	25
3	2	18	0	-2	0	0	4	20	0
4	1	17	-1	-3	3	1	9	15	25
5	3	27	1	7	7	1	49	25	25
Total	10	100	0	0	20	4	114		100

Regression y on X

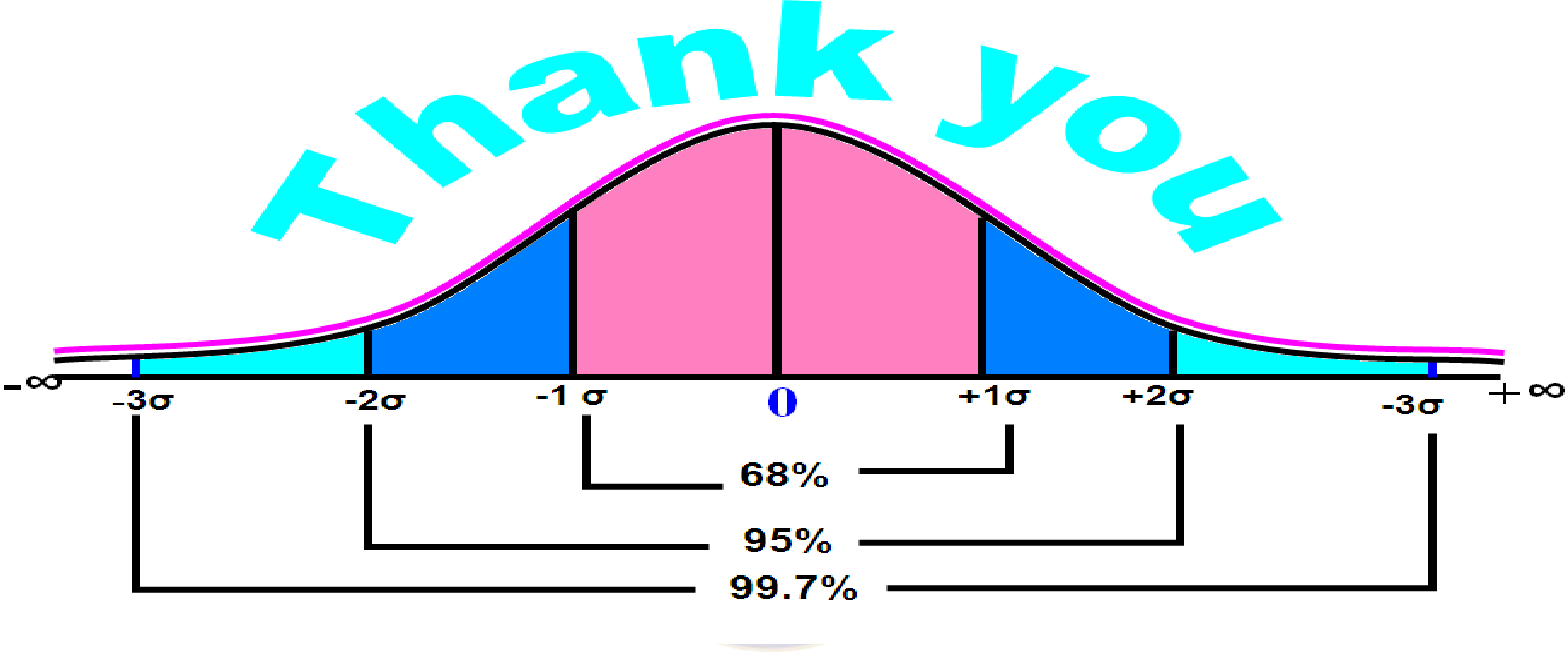
$$\hat{Y} = 10 + 5X$$

Coefficient of Determination

$$r^2 = \frac{SSR}{SST} = \frac{\sum(\hat{Y} - \bar{Y})^2}{\sum(Y - \bar{Y})^2} = \frac{100}{114} 0.8772$$

Interpretation

The regression relationship is very strong; 87.72% of the variability in the number of cars sold can be explained by the linear relationship between the number of TV ads and the number of cars sold



Thank You!